



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

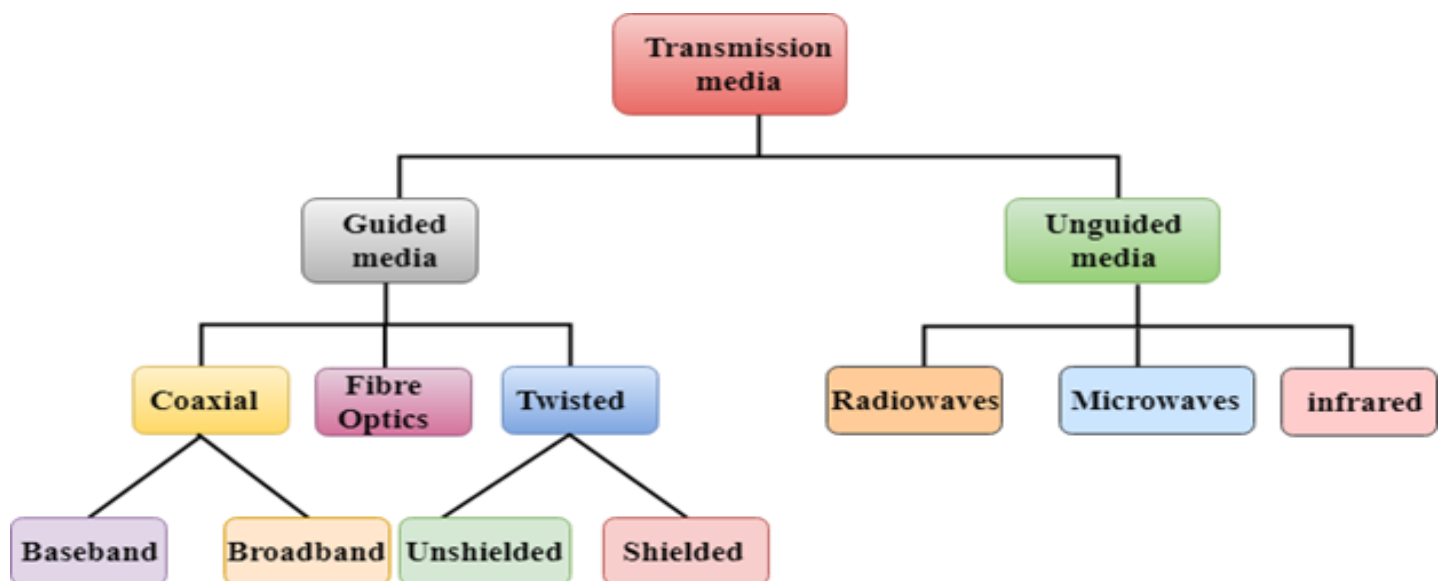
Part -2 (Paper-4)

– Hira Kumar

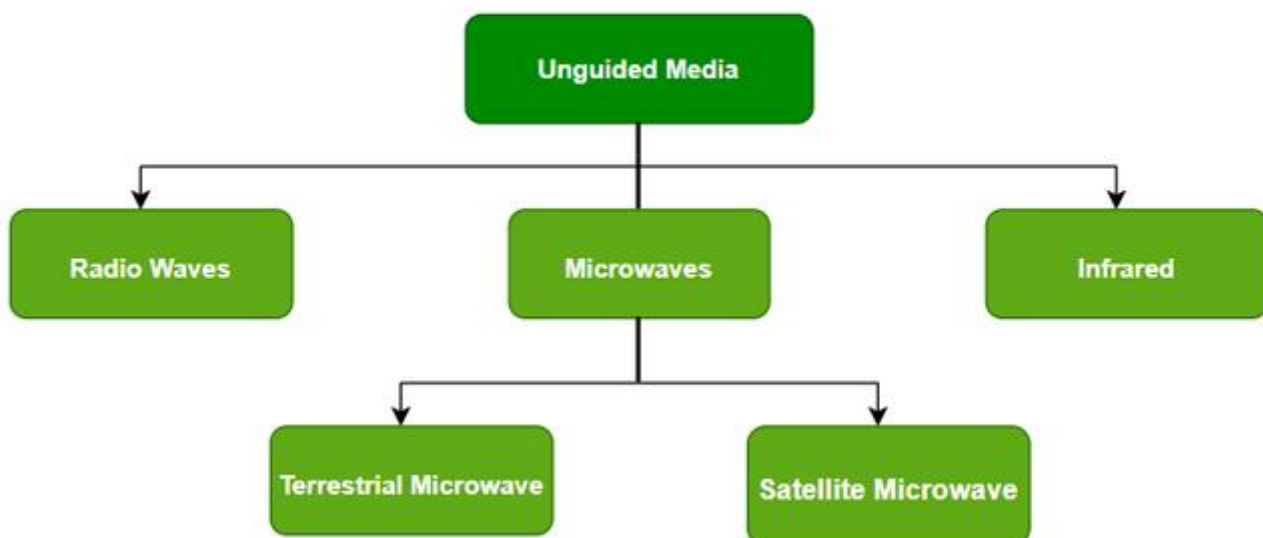
Computer Networking

Data Communication

Types of Transmission Media

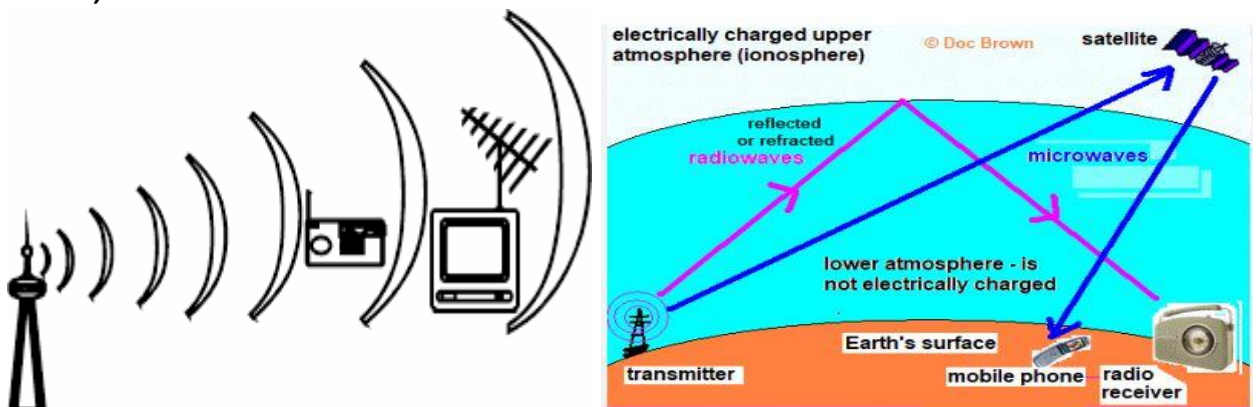


❖ Types of Unguided Media



1. Radiowave Transmission

- Electromagnetic waves with frequencies ranging from 3 kHz to 1 GHz are normally called radio waves.
- Radio wave all propagation (ground, sky and LOS) methods are use.
- Radio waves are **omnidirectional**, i.e., they propagate in all directions from the transmitting antenna.
- A transmitting antenna sends radio waves that can be received by any suitable receiving antenna within the coverage area.
- Radio waves, particularly low and medium-frequency waves, can penetrate walls.
- This is an advantage because AM radio receivers can receive signals inside buildings.
- This is a disadvantage because communication cannot be completely restricted to a specific area (inside or outside a building), and signals may become weak due to interference, rain, dust, and other environmental factors.



➤ Applications/Use

- ✓ The omnidirectional characteristics of radio waves make them useful for multicasting, in which there is one sender but many receivers.
- ✓ AM and FM radio
- ✓ Television
- ✓ Cordless phones and paging.

2. Microwave Transmission

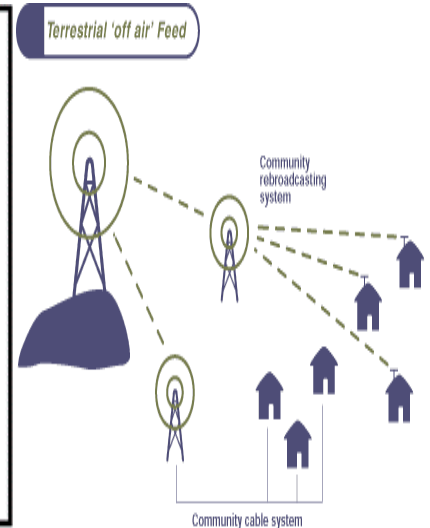
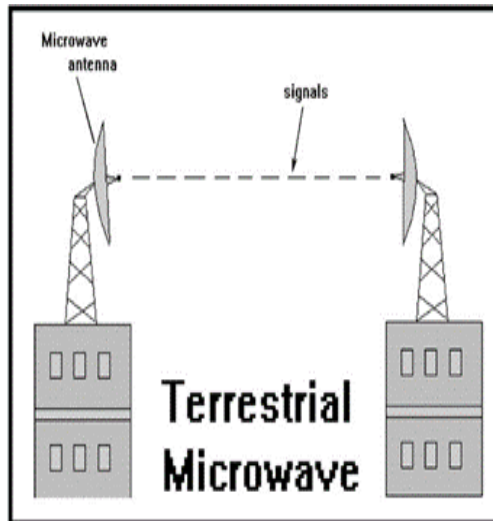
- Microwaves use Line-of-Sight (LOS) propagation.
- Electromagnetic waves having frequencies between 1 GHz and 300 GHz are called Microwaves.

- Microwaves are **unidirectional**.
- Microwaves travel in a straight line and require a clear path between the transmitting and receiving antennas.
- Buildings, trees, hills, or mountains can block/obstruction (रुकावट) microwave signals.
- It can be affected by weather conditions.
- To avoid obstructions, microwave antennas are installed on tall towers, rooftops, or mountain tops.

➤ Microwave transmission is divided into two types-

A. Terrestrial Microwave

- It is used for long-distance communication, such as telephone, television, and data transmission.
- It uses the radio frequency spectrum from 2 GHz to 40 GHz.
- It requires an unobstructed ((बिना रुकावट)) Line-of-Sight (LOS) between the transmitting and receiving antennas.
- Due to the **curvature of the Earth**, microwave repeaters or stations are installed approximately **30 miles (48–50 km)** apart.



➤ Application

- Long-distance telecommunication
- Telephone system
- TV distribution
- Short point-to-point communication links
- Data links between Local Area Networks (LANs)
- Closed-Circuit Television (CCTV)

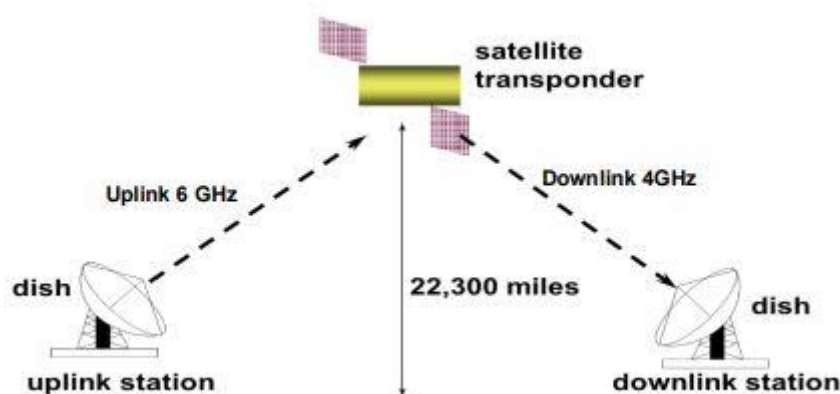
➤ Advantage

- Requires fewer amplifiers or repeaters than coaxial cable, making it suitable for long-distance communication.

B. Satellite microwave / Satellite Communication



- A **communication satellite** acts as a **microwave relay station** in space.
- It is used to connect **two or more Earth stations** (ground-based microwave transmitters and receivers).
- The satellite receives a signal from an Earth station on one frequency band called the **Uplink**, amplifies or regenerates the signal, and retransmits it to another Earth station on a different frequency band called the **Downlink**.
- Satellite communication enables **long-distance point-to-point communication** between widely separated locations on Earth.



Working

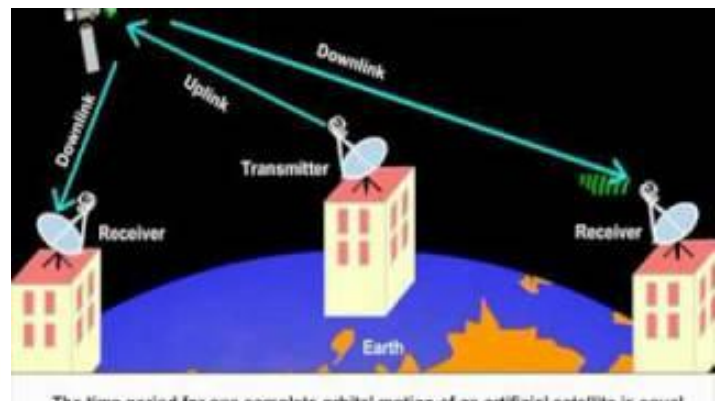
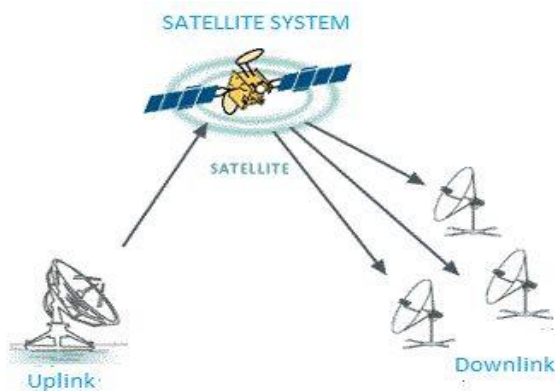
- In the fig the Earth station transmits a microwave signal to the

satellite. This is called the **Uplink (typically around 6 GHz)**.

- The satellite receives, amplifies, and processes the signal.
 - The satellite retransmits the signal back to another Earth station.
- This is called the **Downlink (typically around 4 GHz)**.

➤ **Types of Satellite by there purpose**

- ✓ **Communication Satellite** : Used for TV, telephone, internet, and data communication.
- ✓ **Weather satellite** : Used for weather forecasting and climate monitoring.
- ✓ **Remote- Sensing Satellite** : Used for Earth observation, mapping, agriculture, and resource monitoring.
- ✓ **Scientific Satellite** : Used for space research and scientific experiments.



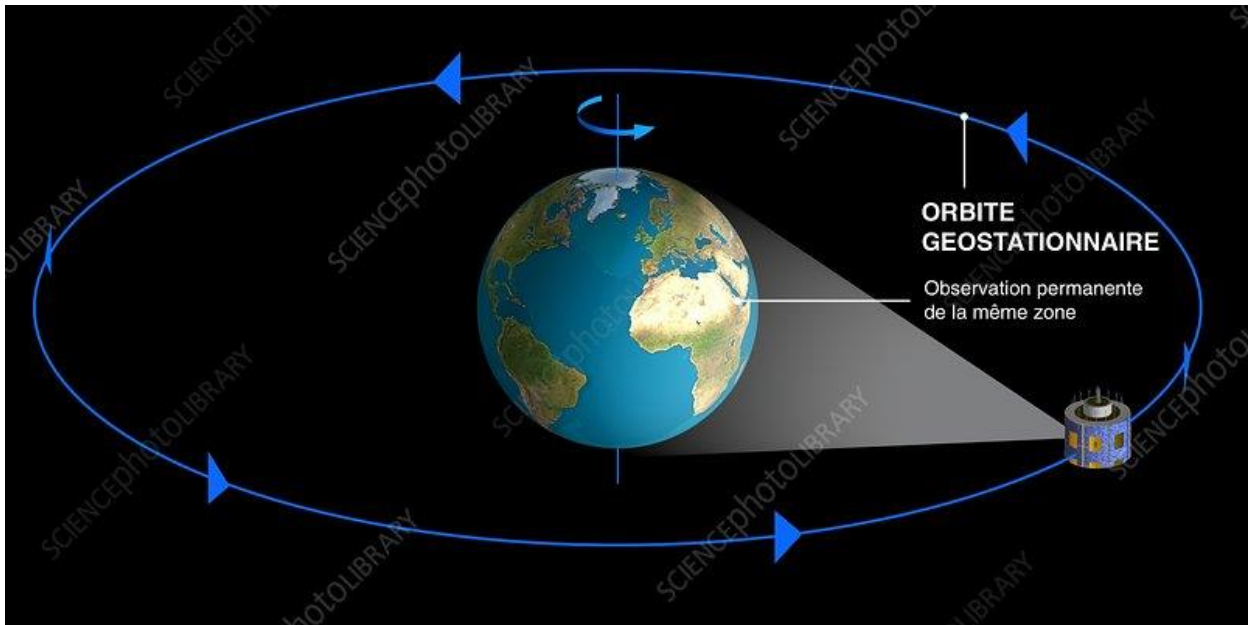
➤ **Application / Use**

- ✓ Television broadcasting
- ✓ Telephone communication
- ✓ Internet services
- ✓ GPS and navigation
- ✓ Weather forecasting
- ✓ Disaster management
- ✓ Military communication
- ✓ Satellite radio
- ✓ Satellite internet access

➤ **Geostationary Earth Orbit (GEO) / GEO Satellite**

- These satellites are in orbit 35,786 km above the earth's surface along the equator(भूमध्य रेखा).
- Objects in Geostationary orbit revolve around the earth at the same speed as the earth rotates.

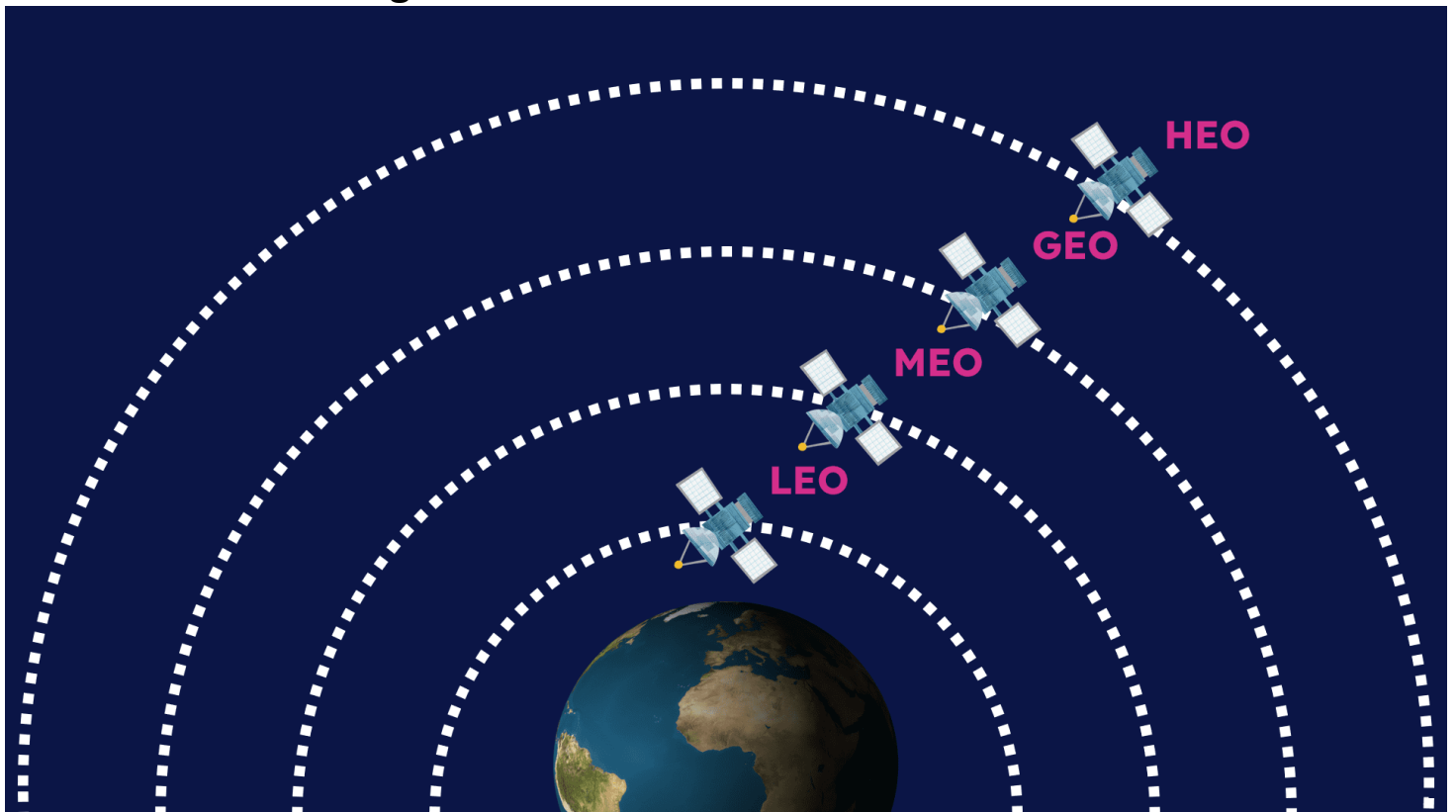
- This means GEO satellites remain in the same position relative to the surface of earth.



Use : TV, DTH (Direct to home), mobile communication,....

❖ Low Earth orbit (LEO / LEO Satellite

- This satellite is a satellite that orbits the earth at a relatively low altitude, generally about 160 to 2000 km above earth's surface.
- It moves around earth very quickly.
- Each satellite covers a smaller area compared with a GEO satellite.
- Many LEO satellites are usually needed for continuous global coverage.
- Starlink uses a large constellation of LEO satellites for satellite internet.





3. Infrared wave transmission

- Infrared waves, with frequencies from 300 GHz to 400 THz.
- Used for short-range communication.
- One important difference between infrared and microwave transmission is that they don't penetrate walls, while infrared across the walls.
- They are relatively directional, cheap, and easy to build.

➤ Applications

- ✓ TV Remote control
- ✓ Guidance in weapon system
- ✓ Wireless keyboards and mouse
- ✓ Infrared therapy.



➤ Difference B/w Radiowave, Microwave & Infrared wave

Sr. No.	Basis	Radiowave	Microwave	Infrared wave
1.	Direction	These are omni-directional in nature.	These are unidirectional in nature.	These are unidirectional in nature.
2.	Penetration (प्रवेश)	At low frequency, they can penetrate through solid objects and walls but high frequency they bounce off the obstacle.	At low frequency, they can penetrate through solid objects and walls. at high frequency, they cannot penetrate.	They cannot penetrate through any solid object and walls.
3.	Frequency range	Frequency range: 3 KHz to 1GHz.	Frequency range: 1 GHz to 300 GHz.	Frequency range: 300 GHz to 400 THz.
4.	Security	These offers poor security.	These offers medium security.	These offers high security.
5.	Attenuation	Attenuation is high.	Attenuation is variable.	Attenuation is low.
6.	Government License	Some frequencies in the radio-waves require government license to use these.	Some frequencies in the microwaves require government license to use these.	There is no need of government license to use these waves.
7.	Usage Cost	Setup and usage Cost is moderate.	Setup and usage Cost is high.	Usage Cost is very less.

Sr. No.	Basis	Radiowave	Microwave	Infrared wave
8.	Communication	These are used in long distance communication.	These are used in long distance communication.	These are not used in long distance communication.

=====HIRA KUMAR=====